

+39 393 006 3841

andreonbonacci95@protonmail.com

linkedin.com/in/andrea-bonacci

github.com/AndreaBonn

andreabonn.github.io

Cagliari, Italy

Andrea Bonacci

DATA ANALYST — AI ENGINEER

PROFILE

Data Analyst and AI Engineer with over 4 years of experience in data-driven consulting for enterprise clients. End-to-end analytics pipelines, BI dashboards and NLP models — from ETL to production deployment with FastAPI and Docker. Current focus on AI Engineering: AI agents, agentic workflows with MCP, tool use and LLM integration in production. Academic tutor for university interns and responsible for technical interviews to hire Data Scientists and AI Engineers. I also delivered an intensive 8-hour university course on AI applied to coding.

PROFESSIONAL EXPERIENCE

LINKALAB S.R.L. · CAGLIARI — part of Lynx Group since 2025 Nov 2021 → present
Consulting company — Internal growth

• Data Analyst & AI Engineer Oct 2025 → present

AI Engineering focus: AI agents and agentic workflows with MCP, tool use, sub-agents and LLM integration (OpenAI, Anthropic). Production AI systems on client projects, including real-time conversational voice AI, agents for document and accounting automation, and RAG for compliance validation against regulatory corpora. Claude Code, Windsurf and Gemini CLI in production. Academic tutor for university interns. Technical interviews for hiring Data Scientists and AI Engineers post-acquisition.

AI Agents MCP LLM Integration Realtime Voice AI RAG Document AI
Claude Code Academic Tutor Technical Hiring

• Data Analyst & Data Scientist Jul 2022 → Oct 2025

End-to-end data pipelines and BI dashboards (Power BI, QuickSight, Tableau, Looker Studio) for enterprise clients. NLP, sentiment analysis, time series. Deployment with FastAPI and Docker on AWS.

Python SQL Power BI QuickSight Tableau NLP FastAPI Docker
AWS S3/Athena

• Data Scientist — contract Apr 2022 → Jun 2022

Post-internship hire. First autonomous data analysis and modeling projects for enterprise clients.

• Data Scientist — curricular internship Nov 2021 → Mar 2022

Master's thesis internship. Development of an Italian Emotion Analysis system for online reputation monitoring.

AT A GLANCE

4+

years of professional
experience

18+

open source projects

DIFFERENTIATORS

AI + BUSINESS

Business degree + Data Science master's = I build AI that solves real problems, from idea to deployment.

AI ENGINEERING

Advanced use of LLMs, AI agents and MCP to build intelligent workflows and automate complex tasks.

UNIVERSITY SPEAKER

AI applied to coding course, University of Cagliari, April 2026 — 8 hours.

MENTORING & HIRING

Academic tutor for university interns. Leading technical interviews for post-acquisition team expansion.

SKILLS

AI & LLM

AI Agents MCP Prompt Eng.
RAG OpenAI API Anthropic API
Tool Use Function Calling
Document AI Human-in-the-Loop
Realtime Voice Sub-agents
Hugging Face Ollama LLM Eval
Agentic Coding Config

DATA SCIENCE

Python SQL NLP
Sentiment Analysis Pandas
Time Series Computer Vision

UNIVERSITY OF CAGLIARI

Design and delivery of two intensive sessions (8 hours total) on “Artificial Intelligence Applied to Coding”. LLM architecture, advanced prompt engineering, agentic coding with CLI tools, Model Context Protocol (MCP) integration and custom Skills creation for AI agents. Hands-on workshops where students built their first AI-assisted development workflows.

AI for Coding Prompt Engineering LLM MCP University Lecture

CLIENT PROJECTS

ON A CONTRACT BASIS FOR AN ENTERPRISE CLIENT · DETAILS ANONYMIZED FOR CONFIDENTIALITY

Real-Time Conversational Voice AI Systems

Design and development of real-time AI voice conversation systems, with automatic evaluation of the interactions. I work with real-time voice AI providers (OpenAI Realtime API, Gemini Live) and streaming transcription (AssemblyAI), multi-provider architectures with fallback and circuit breaker, RAG knowledge bases (embeddings and vector search) via function calling, automatic evaluation with anti-hallucination validation and multilingual prompt engineering. Skills integrable into heterogeneous production stacks.

Realtime Voice AI OpenAI Realtime API Gemini Live AssemblyAI RAG
Function Calling pgvector AWS Bedrock WebSocket Circuit Breaker
Cross-stack Integration

AI Agents for Document & Finance Automation

Design and development of AI agents for document and accounting process automation (e.g. automatic reconciliation of related documents). LLM agents with native function calling and validated structured output (Pydantic), advanced prompt engineering; document AI to extract structured data from heterogeneous documents (XML, PDF, signed files) via parsing and information extraction; deterministic matching with rules, tolerances and confidence scoring; integration with business and ERP systems; human-in-the-loop flows.

AI Agent OpenAI Function Calling Prompt Engineering Document AI
Information Extraction Fuzzy Matching lxml pdfplumber
ERP Integration Python Flask Pydantic

RAG & Document Intelligence for Compliance

Design and development of AI systems for automatic document compliance validation against a regulatory corpus, with source citation and human review. RAG pipelines with hybrid search (BM25 + embeddings) and domain-specific chunking, multi-phase agentic orchestration over LLMs for issues and structured verdicts with regulatory references, validated output and anti-hallucination via mandatory source back-links, human-in-the-loop flows with checkpoint/resume and audit trail. Asynchronous FastAPI backend with WebSocket and document parsing (PDF, DOCX).

RAG Hybrid Search LLM Orchestration Structured Output
Human-in-the-Loop Document AI Haystack OpenAI Embeddings FastAPI
WebSocket Vector Store Async Python

BUSINESS INTELLIGENCE

Power BI QuickSight Tableau
Looker Studio

MLOPS & CLOUD

Docker FastAPI AWS S3 Athena
AWS Bedrock pgvector Flask
GitHub

LANGUAGES

Italian Native
English Fluent

AI & Automation Ongoing

Two GitHub Actions on Marketplace with multi-provider LLM architecture and fallback: [AI PR Reviewer](#) (code review, key rotation) and [AI Changelog Generator](#) (release notes, self-evaluation, multilingual). [Video anonymization](#) with YOLO/CV, [Chrome extension for LLM-powered article/PDF summaries](#), [multilingual TTS](#).

Python

GitHub Actions

GitHub Marketplace

YOLO

Computer Vision

LLM API

Chrome Extension

AI-Powered Web Apps Ongoing

[BetTracker](#) — sports betting tracker with screenshot OCR, automatic result resolution, AI chat and advanced statistics. PWA with offline, OAuth and cloud sync.

React

TypeScript

Gemini AI

Firebase

PWA

Tailwind CSS

Full-Stack Web Apps Ongoing

PWAs with cloud sync and offline: [finance](#), [health](#), [shared expenses](#). End-to-end ownership.

JavaScript

PWA

Cloud Sync

Responsive UI

Developer Tools & CLI Ongoing

[Claude Code Tools Explorer](#) — TUI for config inspection (Textual/Rich). [Command Quiver](#) — desktop prompt & shell manager (GTK4, SQLite). [Telegram bot for remote multi-PC shell](#) (zero infrastructure, heartbeat, audit log, systemd), [file manager extensions](#), [clipboard image CLI](#), [audio renaming via AcoustID](#), and [smart clock-in notifications](#).

Python

JavaScript

asyncio

Textual/Rich

Telegram Bot

Shell

Linux

systemd

CLI

Chrome Extension

— EDUCATION

Master’s Degree — Data Science & Business Informatics 2022

University of Pisa

Thesis: Italian Emotion Analysis for online reputation monitoring

Bachelor’s Degree — Business Administration 2018

University of Pisa